



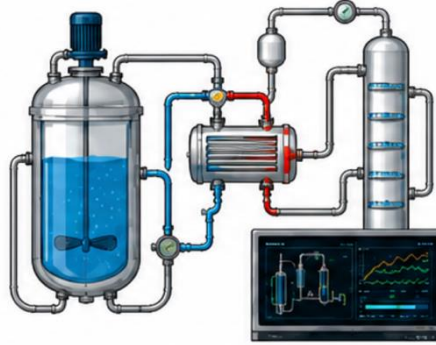
Safety-Aware Performance Boosting for Constrained Nonlinear Systems

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Motivation



- Fast motion
- Collision avoidance



- Tight constraints
- Economic performance



- Reliability
- Transient performance



Safety



Stability



Performance

In many control applications, we need safety, stability, and high performance – all at once.

Problem setting

- **Goal: optimize performance subject to hard safety and closed-loop stability.**

$$\min_K \mathbb{E}_{x_0 \sim \mathcal{D}} \sum_{t=0}^{\infty} \gamma^t l(x_t, u_t)$$

$$x_{t+1} = f(x_t, u_t)$$

$$u_t = K(x)$$

$$x_t \in \mathcal{X}, \quad u_t \in \mathcal{U}$$

$$x_t \in \ell_2, \quad u_t \in \ell_2$$



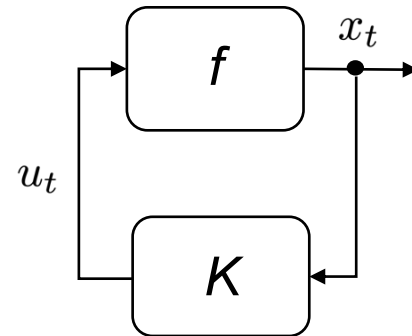
Safety



Performance

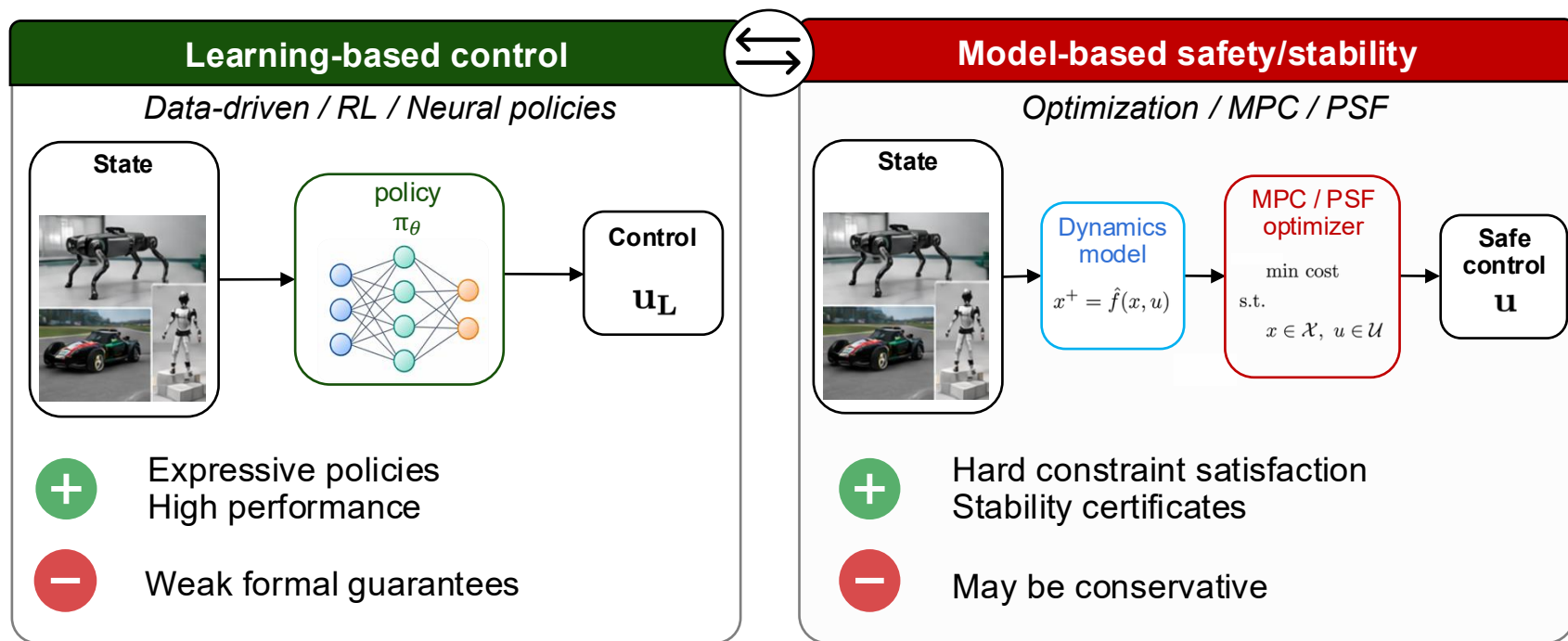


Stability



Can we improve performance without sacrificing hard safety and closed-loop stability?

Learning vs. model-based control



Goal: learning-based performance with model-based guarantees.

- A PSF projects the learning input onto a safe admissible set.

Online optimization at each time t

$$\min_{u_{\cdot|t}} \|u_{0|t} - u_{L,t}\|^2$$

s.t. $x_{0|t} = x(0)$

$$x_{i+1|t} = f(x_{i|t}, u_{i|t}), \quad \forall i \in \mathbb{N}_{[0, N-1]}$$

$$x_{i|t} \in \mathcal{X}, \quad \forall i \in \mathbb{N}_{[0, N]}$$

$$u_{i|t} \in \mathcal{U}, \quad \forall i \in \mathbb{N}_{[0, N-1]}$$

Stability constraint $J_{i+1|t} \leq J_{i|t} - (1 - \bar{\rho})s_{i|t}$

Lyapunov decrease with $0 < \bar{\rho} < 1$

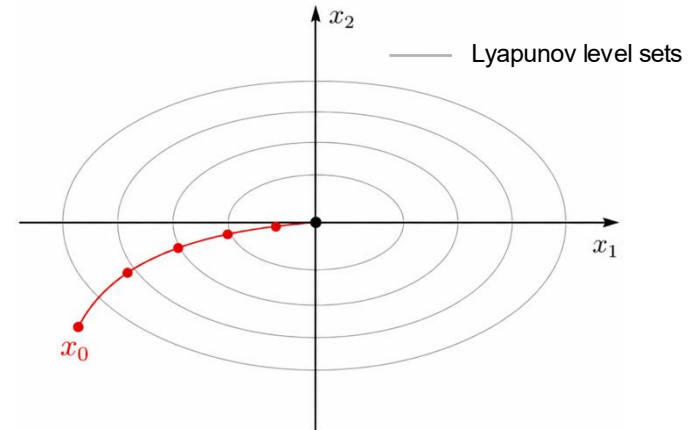
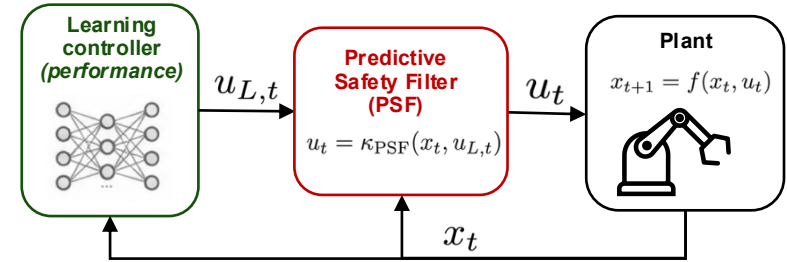


If initially feasible

$$x_t \in \mathcal{X}, \quad u_t \in \mathcal{U}$$



Stability via monotonic decrease of the certificate



Limitation: Fixed Decrease Can Reject Safe Actions

Projection viewpoint

$$u_t = \text{Proj}_{U_t(x_t, J_{t-1}^*, \bar{\rho})}(u_{L,t})$$

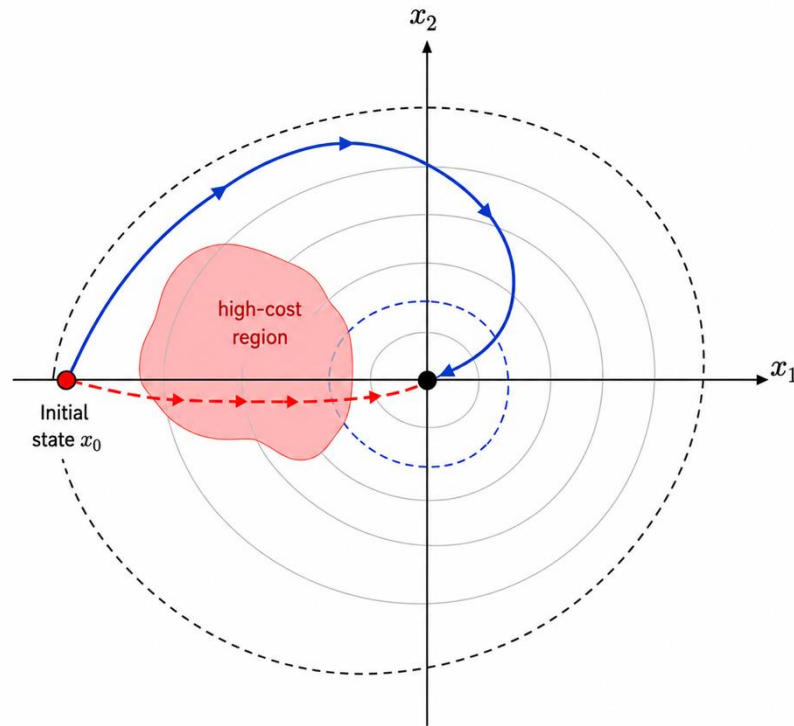
$$U_t = \{\text{safe inputs}\} \cap \{\text{certificate-decreasing input}\}$$

- 1  PSF **projects** the proposed input onto the admissible set
- 2  Fixed decrease forces J_t^* to decrease
- 3  As J_{t-1}^* shrinks, U_t becomes more restrictive



Safe inputs may be **rejected** because they are outside the shrinking projection set.

State-space intuition



Fixed decrease guarantees stability, but may exclude safe performance-improving transient behaviors.

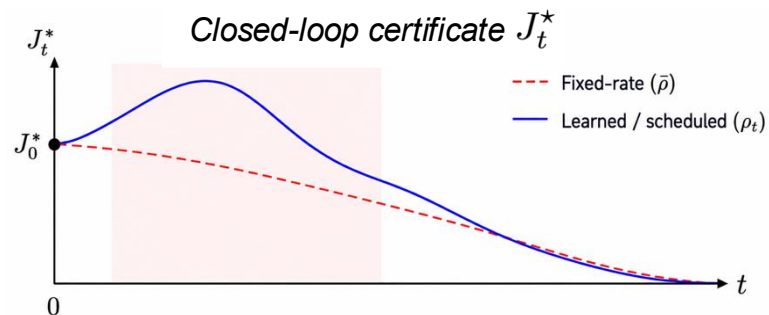
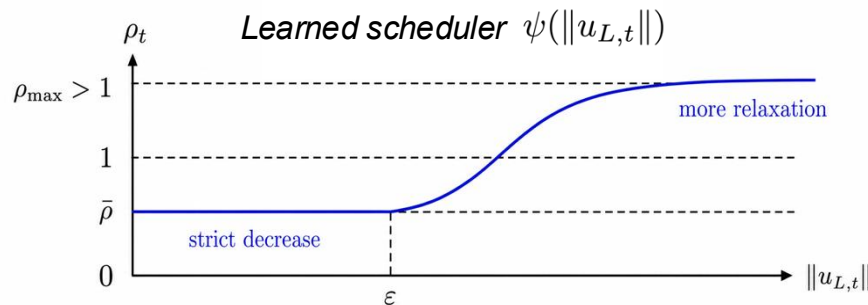
Main idea: learn when to relax the decrease

Scheduled PSF:

$$J_t^* \leq J_{t-1}^* - (1 - \rho_t) s_{t-1} \quad \rho_t = \psi(\|u_{L,t}\|)$$

learned scheduler
(performance-aware
scheduler)

- Large $\mathbf{u}_L \rightarrow$ larger $\rho_t \rightarrow$ more transient freedom
- $\mathbf{u}_L \in \ell_2 \iff \|\mathbf{u}_L\|_{\ell_2}^2 := \sum_{t=0}^{\infty} \|u_{L,t}\|^2 < \infty \implies \|u_{L,t}\| \rightarrow 0$
- Strict Lyapunov decrease is recovered eventually



We do not need monotone decrease at every instant; we need to recover strict decrease eventually.

Separation of roles

- One learned signal $\mathbf{u}_L$ has two roles: it proposes a performance action and schedules the Lyapunov decrease rate

Separation of roles



PB controller proposes $\mathbf{u}_L$

$$\mathbf{u}_L = \mathcal{M}_\theta(\mathbf{x}, \mathbf{x}^{(0)})$$



$$\rho_t = \psi(\|\mathbf{u}_{L,t}\|)$$

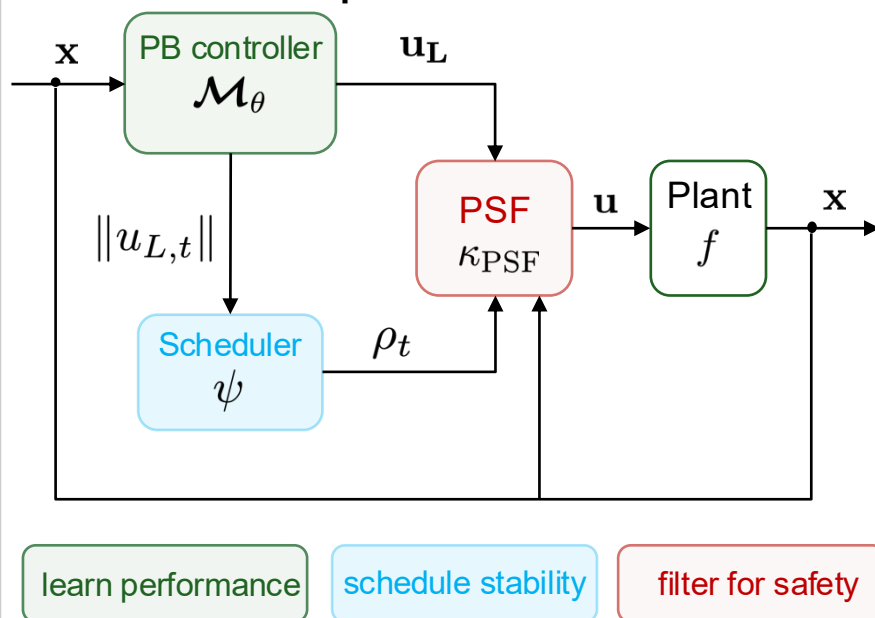
If $\mathbf{u}_L \in \ell_2$, then ρ_t eventually returns to $\bar{\rho} < 1$



PSF enforces hard constraints

$$\mathbf{u}_t = \kappa_{\text{PSF}}(\mathbf{x}_t, \mathbf{u}_{L,t}, \rho_t, \mathbf{J}_{t-1}^*)$$

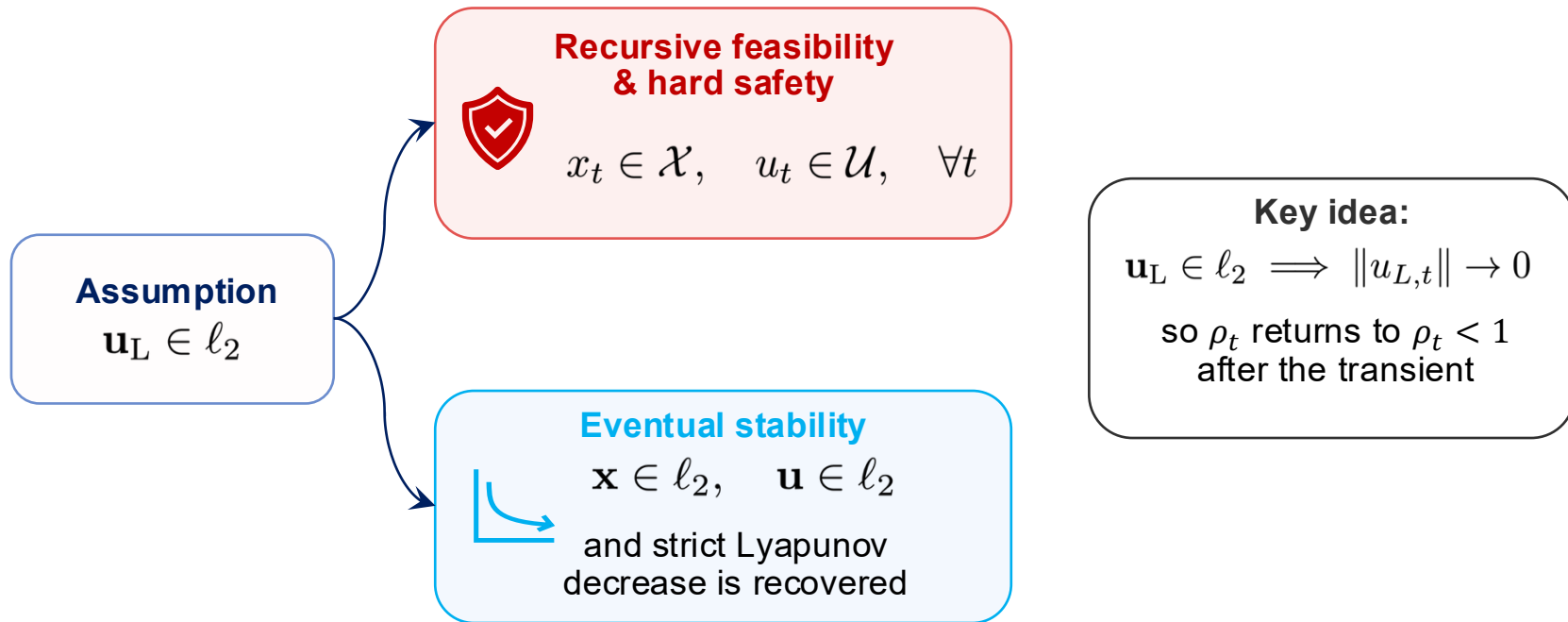
Closed-loop interconnection



The PB controller drives performance; the PSF enforces safety; the scheduler recovers strict Lyapunov decrease.

Main result 1: Safety and eventual stability

- One learned signal $\mathbf{u}_L$ has two roles: it proposes a performance action and schedules the Lyapunov decrease rate



Safe by design, eventual stability after the transient

Main result 2: Trajectory-set expansion

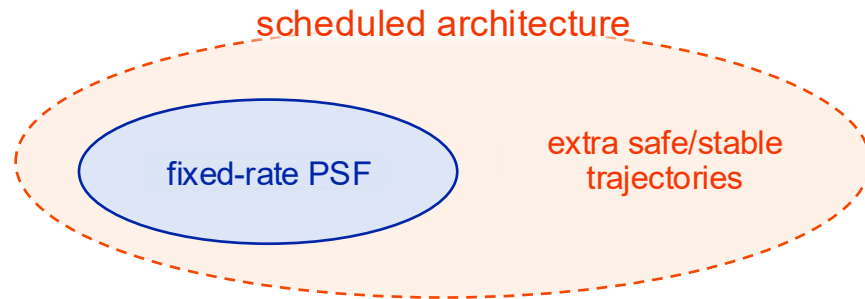
- The scheduled PSF strictly enlarges the set of safe and stable closed-loop behaviours

$$\mathcal{B}(\text{PSF}_{fix}(\bar{\rho})) \subseteq \mathcal{B}(\text{PSF}_{sch}(\psi))$$

Strict enlargement



When the Lyapunov constraint is active, one relaxed feasible action can generate a trajectory unattainable by any fixed-rate PSF



When $\rho_t > \bar{\rho}$, the admissible input set $U_t(\rho_t, J_{t-1}^*)$ enlarges

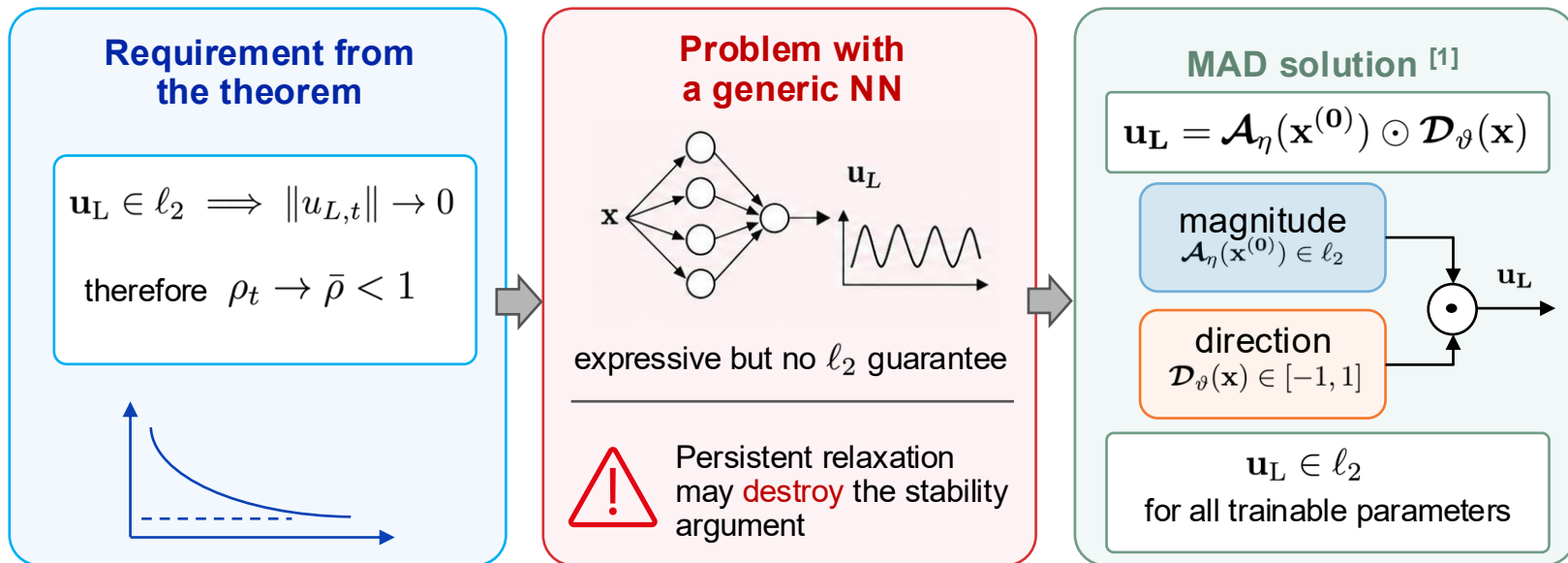


The scheduled PSF can **reproduce** the fixed-rate PSF, but can also **realize trajectories that the fixed-rate PSF cannot**.

Safety by design, with a strictly larger set of safe and stable behaviors

PB controller parametrization

- The PB controller must be expressive, but its output must vanish eventually.



MAD separates **how much** to act from **where** to act

Expressive performance signal, with ℓ_2 property guaranteed by construction

Actor–critic training around the PSF

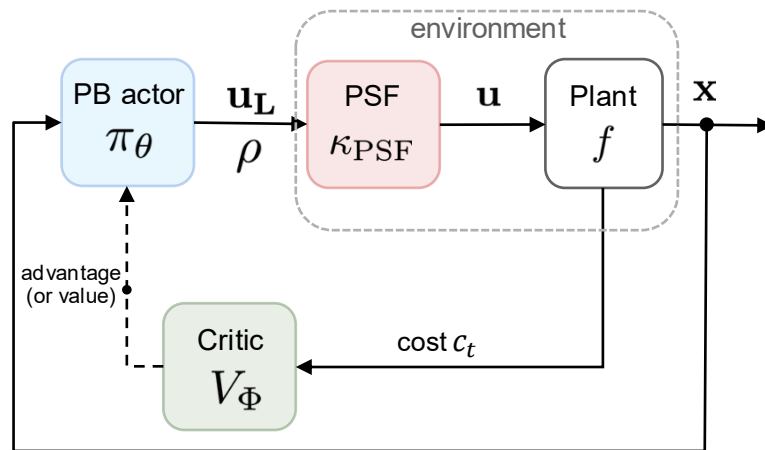
Challenge

The PSF solution map can be nonsmooth.

Idea

Train the PB controller using actor–critic, treating PSF + plant as a black-box environment.

- Actor proposes $\mathbf{u}_L$
- PSF filters before the plant.
- Critic learns from safe closed-loop rollouts.



Safe exploration



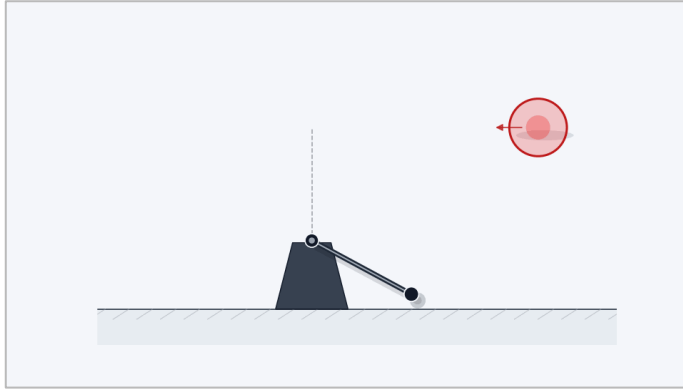
All exploratory actions are filtered by the PSF before reaching the plant

Safety is preserved during training because every exploratory action is filtered.

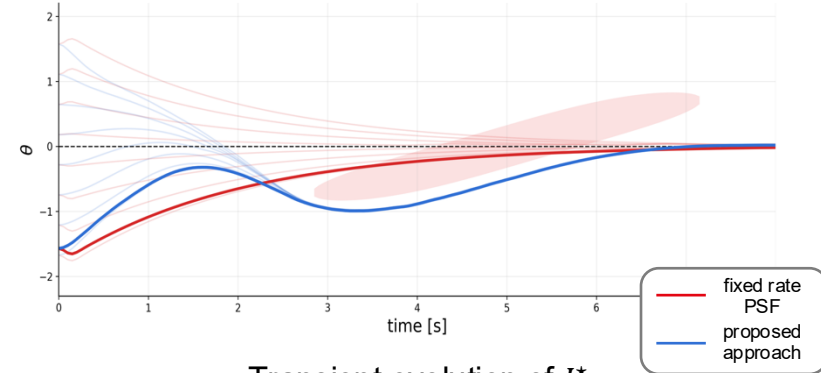
Numerical example: pendulum with moving obstacle

Task

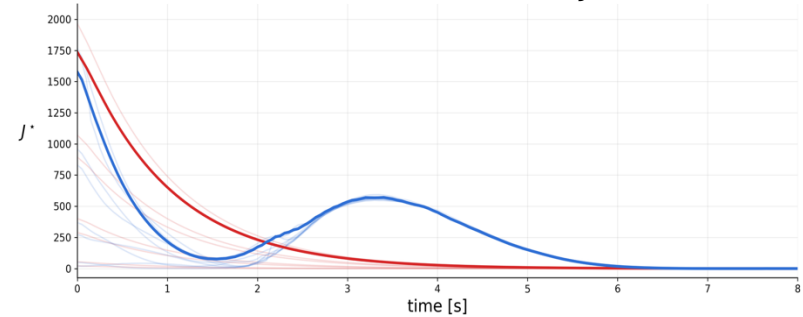
- stabilize the pendulum upright $x = [\theta, \dot{\theta}]^\top$
- avoid a moving obstacle
- a transient detour is required
- the obstacle blocks the direct stabilizing path



Time evolution of θ



Transient evolution of J^*



The scheduled PSF allows a transient increase of J^* , avoids the obstacle, and then recovers strict decrease.

Conclusions

- We can enlarge the set of safe and stable behaviors by scheduling Lyapunov decrease.



Safety
PSF enforces hard constraints

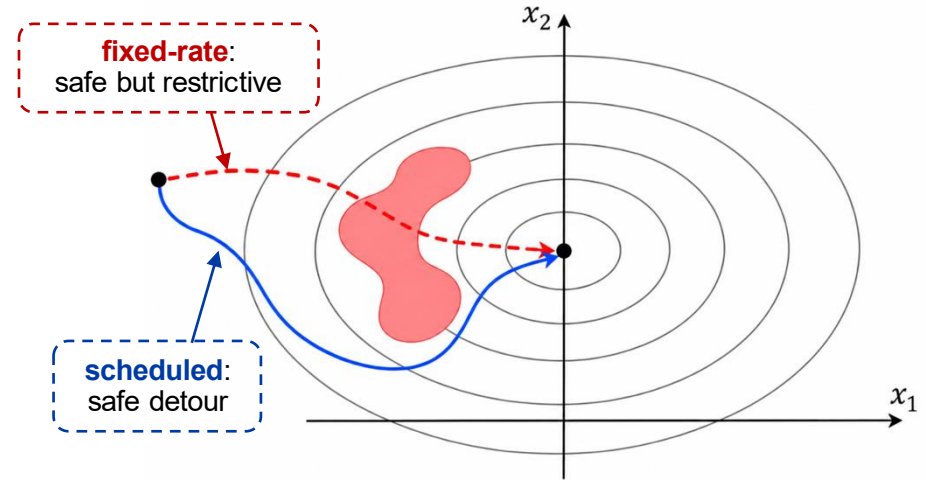


Stability
Strict decrease is recovered when u_L vanishes



Performance
PB controller enables useful transient detours

- Fixed Lyapunov decrease can be overly conservative.
- The learned performance signal schedules temporary relaxation
- Result: hard **safety** + eventual **stability** + **larger behavior set**



Preprint



Learn performance without giving up safety and stability